

# Kaden Seto

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## EDUCATION

### University of Toronto

*Bachelor of Applied Science & Engineering (BASc) in Engineering Science + PEY Co-op*

Toronto, ON

Sep 2024 - 2029

## TECHNICAL SKILLS

**Languages:** Python, C#, C/C++, JavaScript/TypeScript, Java, HTML/CSS, MATLAB

**Developer Tools:** Git, Visual Studio Code, Unity, Node.js, Android Studio

**Libraries:** Unity ML-Agents, PyTorch, Gymnasium & SB3, Keras, NumPy, pandas, OpenCV, scikit-learn

## EXPERIENCE

### University of Toronto Machine Intelligence Student Team | [Website](#)

Toronto, ON

*Vice President, Academics (May 2025 - present)*

Nov 2024 - present

- Management of the Academics department and initiatives, encompassing **12 directors and leads** across **5 Academic workshops** throughout the year, the 2025-26 **AI<sup>2</sup> RL tournament**, and the **ML Fundamentals** and **Paper Readings** programs supporting **3000+ students** (North America's largest undergraduate ML/AI team)

*Operations Lead (Feb 2025 - Mar 2025)*

- Led the Application Review committee of **25+ members** to review applications of **1300+ applicants** for the **GenAI Genesis 2025 Hackathon (Canada's Largest AI Hackathon)**

*Reinforcement Learning Academics Team Lead (Nov 2024 - Mar 2025) | [AI<sup>2</sup> Website](#) | [Github](#)*

- Directed a team of developers to design UTMIST AI<sup>2</sup>, UofT's largest **RL tournament**, for **400+ participants**
- Developed a custom **multi-agent RL self-play training process** that supports custom PyTorch and SB3 agent implementations and a custom platform-fighting **game environment** using **Pygame** and **Pymunk**
- **First-authored** a paper presented and submitted at CUCAI (Canada's largest undergrad AI conference)

### BMO Financial Group

Toronto, ON

*Software Developer Intern*

May 2025 - Aug 2025

- Developed automation scripts for environment syncing and AWS CDK Stack deployment, **reducing process time by 2+ hours**

## PROJECTS

### NeuroStrike | [Github](#)

May 2025 - Aug 2025

- Engineered a **Reinforcement Learning (RL) framework** for training martial arts AI agents to spar in a **custom-made 3D simulation/environment** using **self-play recurrent PPO (LSTM + PPO)**.
- Engineered **hyperparameters, architectures, reward functions, and observation spaces** to optimize policy behaviour for sampling effective movement and attack actions.

### Aegis | [Github](#)

Jan 2025 - Apr 2025

- Engineered a **Cyclist Detection and Arduino flashing LED system** aimed to prevent right-hooking cyclist incidents in Toronto, collaborating with the Advocacy for Respect for Cyclists group (ARC) to meet client needs
- Trained a **custom object detection model** using YOLOv11 architecture, achieving a **mAP50-95 of 84.6%**, average **50ms inference time** and detections up to **3.2m** on a standard phone camera
- Explored **super-resolution GAN (SRGAN)** models, implementing the **Swift-SRGAN model in PyTorch** aimed to improve detection through real-time upscaling of captured frames

### QuantNet | [Github](#)

Oct 2024 - Feb 2025

- Implemented a hybrid **LSTM & MLP neural network framework from scratch** with **0 external dependencies** (i.e. methods such as **back-propagation** and **Adam** made from scratch) for investment analysis
- Implemented a **feature engineering framework from scratch** by **researching key investment analysis indicators** to enhance model quality and performance (e.g. features such as RSI, ADX)

### OtakuNet | [Github](#)

Aug 2024 - Sep 2024

- Trained a **content-based recommender system** to recommend animes based on the genres users rate by building a **neural network framework from scratch** for a custom dual network architecture (using only **NumPy**)
- Developed **data engineering techniques** such as **Bayesian Ratings** and **normalization** from scratch and applied them to the dataset. Used **Pandas** to organize data and perform data preprocessing tasks